Houssam Eddine Ben Kheder

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Profile

Data science enthusiast with strong foundations in mathematics, statistics, and programming. Experienced in Python, SQL, and data visualization, with a focus on predictive modeling and strategic analysis. Committed to transforming complex data into actionable insights to support informed decision-making.

Education

ESPRIT: Private Engineering School of Technologies **Sep 2021 - Present**
Computer Engineering, Computer Engineering

Wafa School el Gazzella city, Ariana, Tunis **Sep 2017 - Jun 2020**
Baccalaureate Diploma, Mathematics

Projects

AI-Powered Internship GeoMarketing Project: Web Scraping and Agent-Based Simulation **Jun 2025 - Sep 2025**
Esprit, VALUE Enterprise Intern

- The goal of this project is to simulate the behavior of retail and corporate clients of a Tunisian bank using AI-based agent simulations. This simulation will support the geomarketing department's mission to enhance the bank's attractiveness and competitiveness by modeling client responses to strategic changes in products, services, geography, digital offerings, and external socio-economic factors.
- Developed an AI-powered GeoMarketing system targeting professionals (doctors, advocates, architects) in Tunisia by web scraping their geographic and professional data.
- Utilized CTGAN and TVAE to synthesize realistic individual profiles for enhanced dataset robustness.
- Implemented agent-based AI models to simulate socio-economic behaviors using Mesa for agent-based modeling, incorporating event injection to observe behavioral responses.
- Conducted A/B testing to evaluate marketing solutions and visualized results using Dash and Plotly for interactive insights.

DATA SCIENCE PROJECT: AI-Powered Crypto Fund Due Diligence and Risk Assessment **Jan 2025 - May 2025**
ESPRIT, Value Enterprise

- Developed an Al-driven system for conducting due diligence on investment funds, utilizing Machine Learning and Generative AI to generate interactive risk assessments.
- Applied anomaly detection, fraud detection models, and sentiment analysis to evaluate risk exposure in financial assets.
- Created an automated PPTX report generator using Python to provide real-time insights for investors.
- Designed and implemented ML pipelines in compliance with Microsoft's TDSP (Team Data Science Process) framework.
- Incorporated inductive prompting and reverse engineering using RAG and LLM to enhance risk assessment processes and improve decision-making insights.
- Leveraged a DGX workstation for efficient preprocessing, training and deployment for large-scale financial datasets.

Machine Learning Project on Churn Retention and Prediction **Sep 2024 - Dec 2024**
Esprit

- Developed a machine learning model to predict customer churn for a telecommunications company. The project involved data preprocessing, feature engineering, and the implementation of various machine learning algorithms such as Logistic Regression, Random Forest, Gradient Boosting, PCA, and Clusters.
- Utilized Python libraries including Pandas, NumPy, Scikit-learn, and Matplotlib for data analysis and visualization.
- Achieved an accuracy of 99% in predicting churn, outperforming scientific papers, which helped the company in retaining customers by implementing targeted retention strategies.

PIDEV Project on Healthcare using JAVA/SYMFONY/PHP **Jan 2024 - May 2024**
Esprit

- Developed an application to simplify doctor appointments, prescription management, and pharmacy localization. Features include SMS messaging, MAP integration for pharmacy search, and user-friendly interfaces for patients and doctors.

Skills

- **Hard Skills**: NLP, LLM, RAG, ML, PowerBI, Dash Plotly, Project Management, Mathematics, .NET, Python, C++, JAVA, SQL, postgreSQL, MongoDB, Probability, Linux, UML, Statistics, R studio, Data Science, Data Analytics
- **Soft Skills**: Presentation, Planning, Creative Problem-Solving, Teamwork, Active Listening, Adaptability, Analytical Thinking
- **Interests**: Playing Football, Swimming, Gaming, Music, Travelling, Anime

Languages

- Arabic (Native)
- French (Good)
- English (Good)